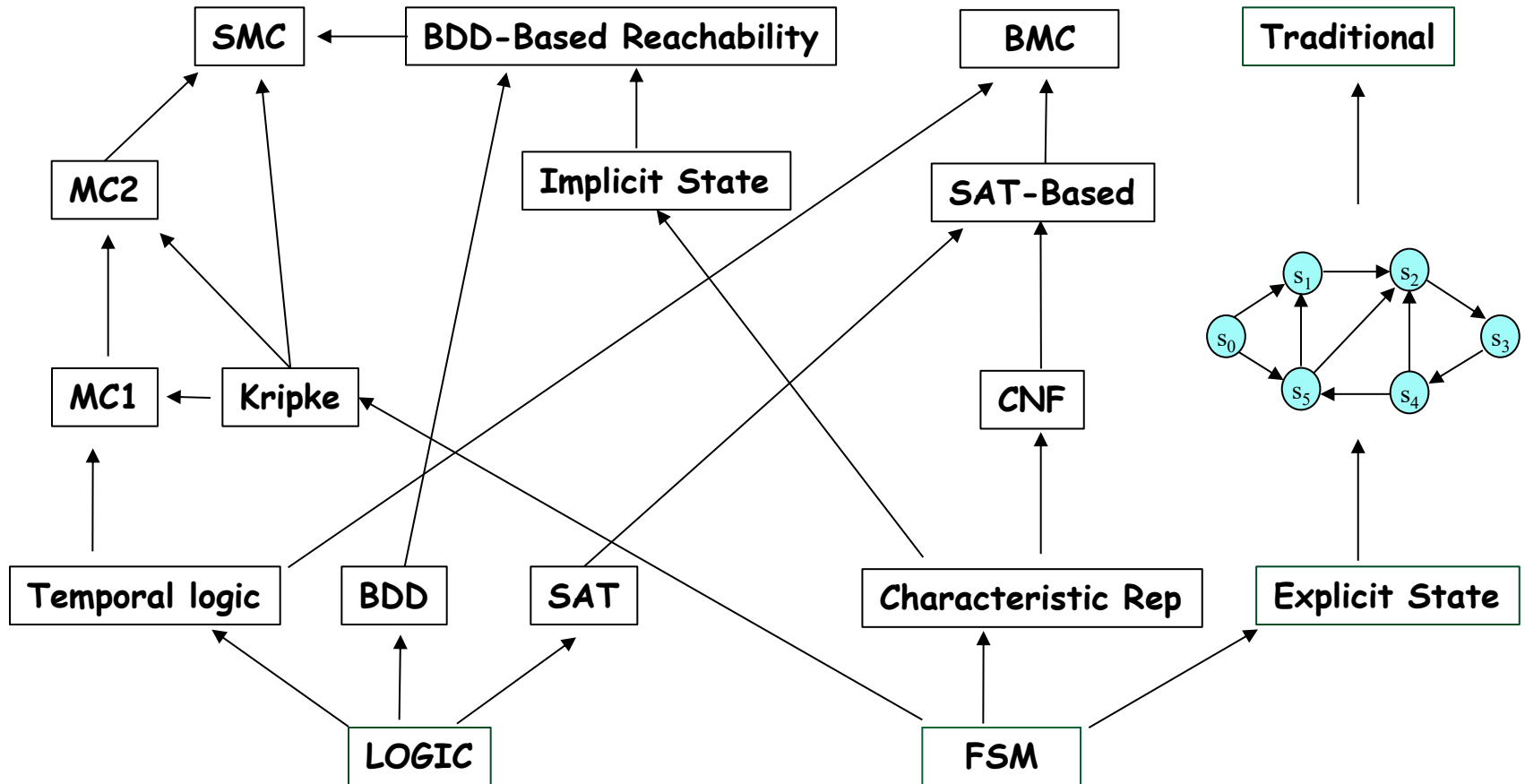
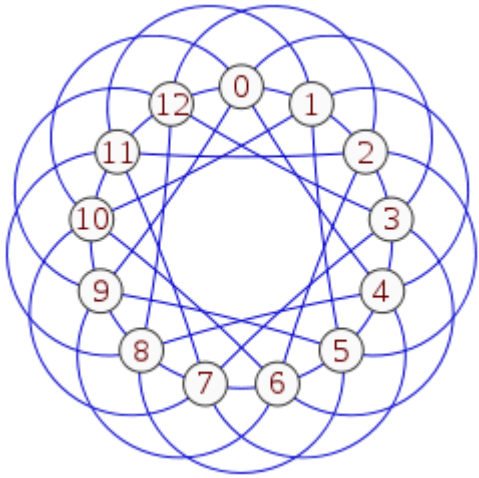


CTL Fixpoint Model Checking (Simplified)

A Road Map



Model Checking Problem

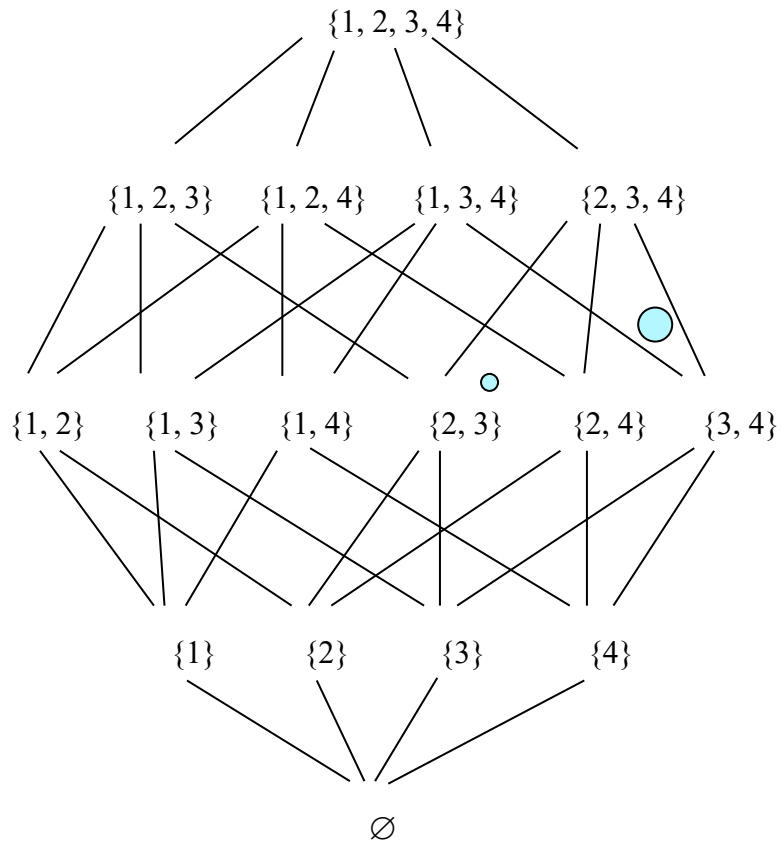


$M, \{?????\} \models \text{Property}$



Subset of States

Set of All the Subsets

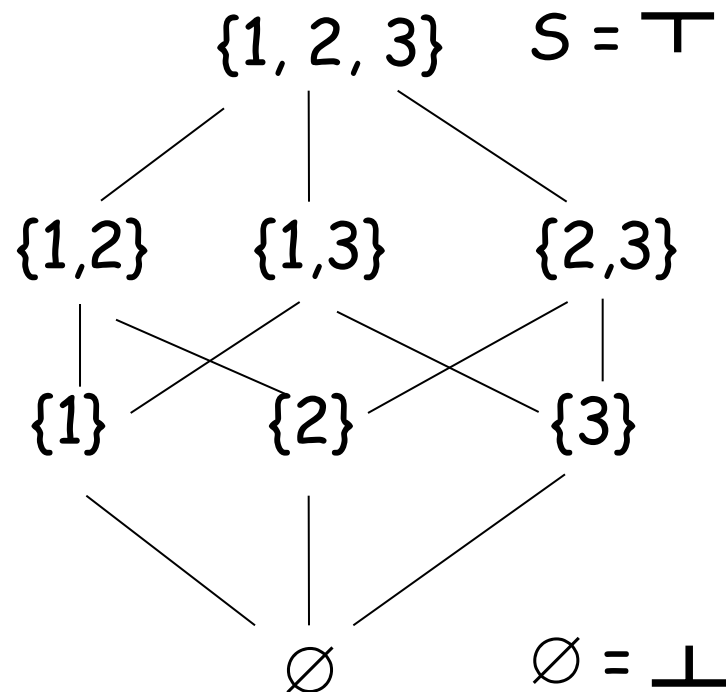


Which Subset???

$S = \{1, 2, 3, 4\}$

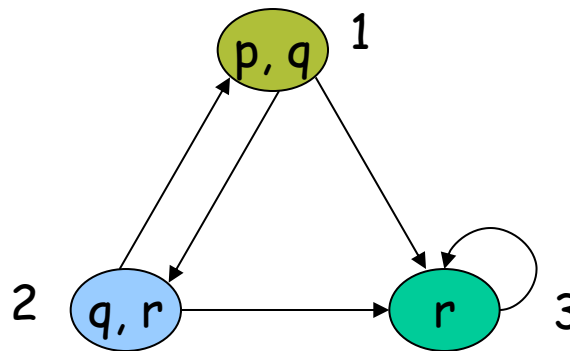
Power Set 2^S of S

- For example: $S = \{1, 2, 3\}$



CTL MC = Find Subsets

- CTL Model Checking is to build an one-to-one mapping between logic formula and a subset of states.



$M, \{2\} \models EXp$

$M, \{1, 2\} \models EFp$

$M, \{2, 3\} \models EGr$

.....



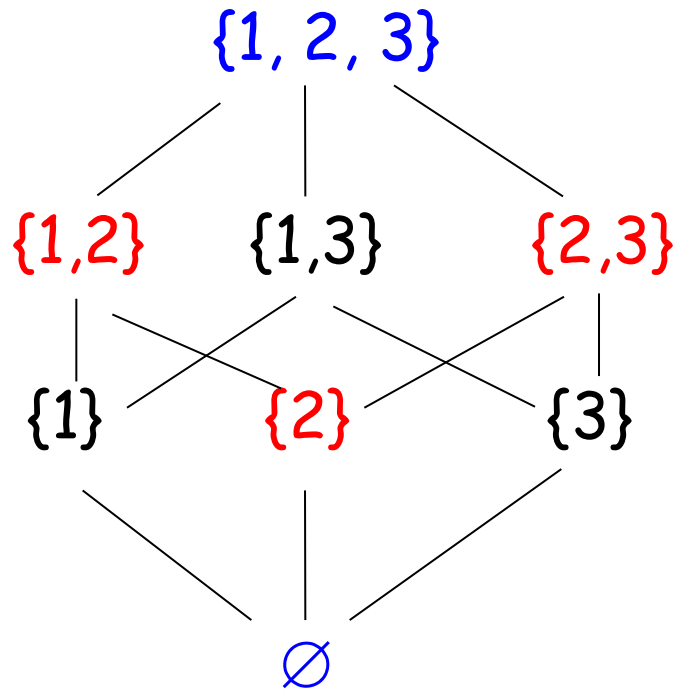
$EXp \Leftrightarrow \{2\}$

$EFp \Leftrightarrow \{1, 2\}$

$EGr \Leftrightarrow \{2, 3\}$

.....

Logic $\Leftrightarrow$ Sets



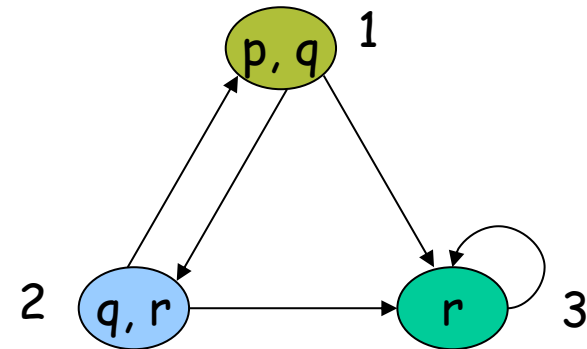
$EXp \Leftrightarrow \{2\}$

$EFp \Leftrightarrow \{1, 2\}$

$EGr \Leftrightarrow \{2, 3\}$

$F \Leftrightarrow \emptyset$

$T \Leftrightarrow \{1, 2, 3\}$



Fixpoint Characterization of CTL

- Identify each CTL formula f with a set of states S_f
 - $f = \{s \mid f \text{ is true on } s \in S\}$.

- Any formula f
- False
- True

$\Leftrightarrow$

$\Leftrightarrow$

$\Leftrightarrow$

a set S_f of states where f is true
the empty set $\emptyset$ ($\perp$)
the total set of states S ($\top$)

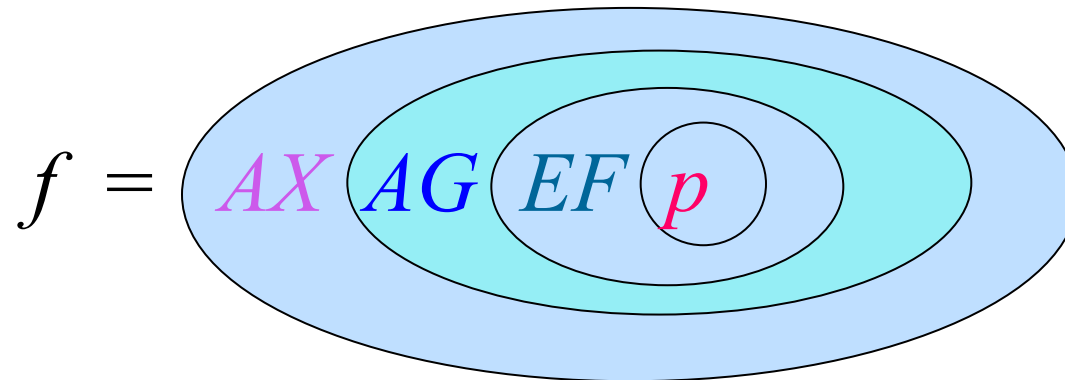
LOGIC

SETS

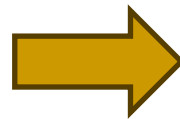
Dual Nature: Logic and Set

- Identify each CTL formula f with a set of states S_f
 - $f = \{s \mid f \text{ is true on } s \in S\} = S_f$.
 - Any formula $f \Leftrightarrow$ a set S_f of states where f is true.
 - f has a dual structure:
 - For a state s where f is true, we have:
 - 1) as logic, $s \models f$.
 - 2) as set, $s \in f$ means $s \in S_f$

CTL Model Checking



- $\{\text{????}\} \models \text{AF}p$
- $\{\text{????}\} \models \text{EF}p$
- $\{\text{????}\} \models \text{AG}p$
- $\{\text{????}\} \models \text{EG}p$
- $\{\text{????}\} \models \text{A}(p\text{U}q)$
- $\{\text{????}\} \models \text{E}(p\text{U}q)$

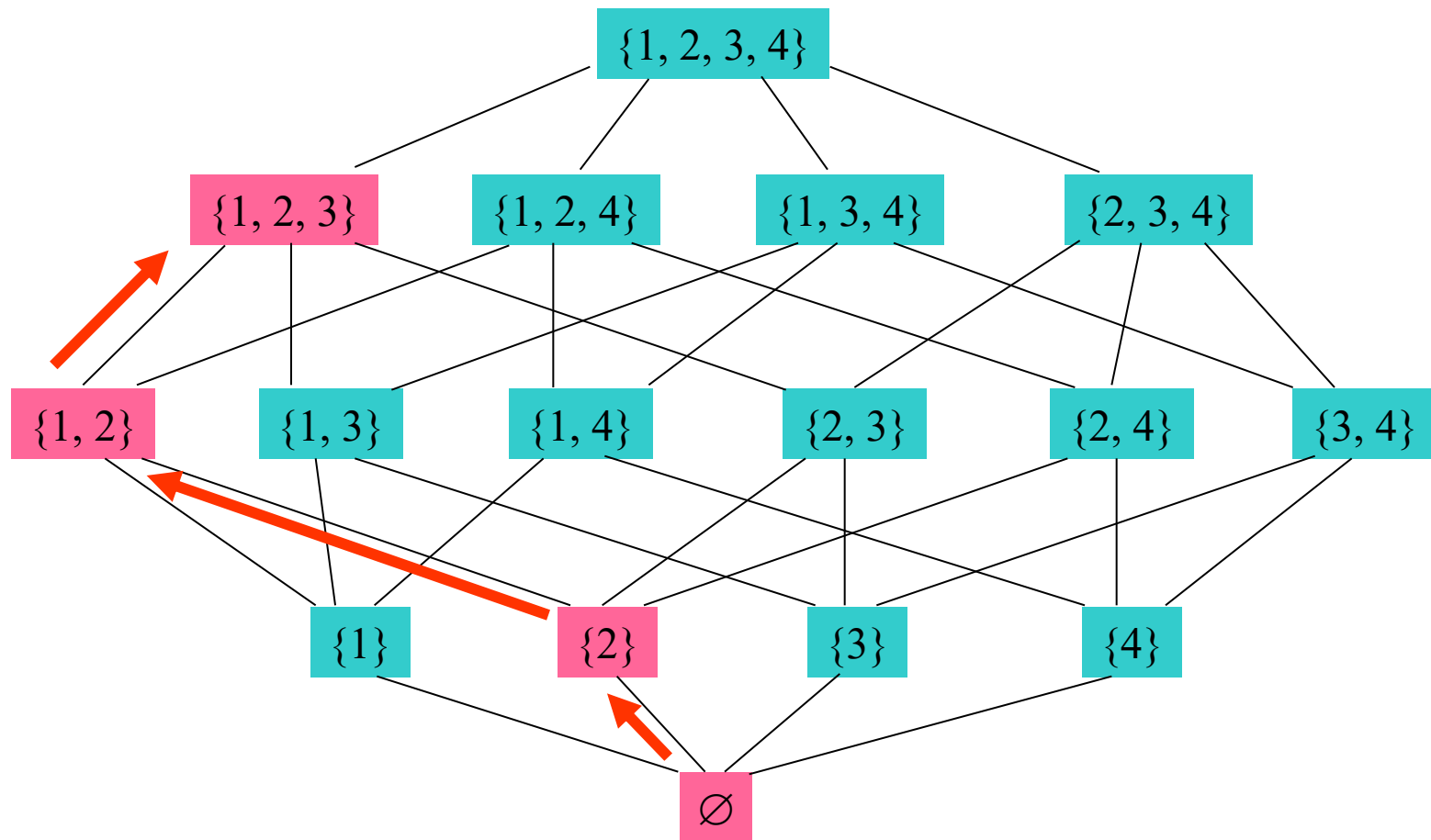


- $\text{AF}p = \{\text{????}\}$
- $\text{EF}p = \{\text{????}\}$
- $\text{AG}p = \{\text{????}\}$
- $\text{EG}p = \{\text{????}\}$
- $\text{A}(p\text{U}q) = \{\text{????}\}$
- $\text{E}(p\text{U}q) = \{\text{????}\}$

CTL Model Checking

- $AFp = \{????\}$
- $EFp = \{????\}$
- $AGp = \{????\}$
- $EGp = \{????\}$
- $A(pUq) = \{????\}$
- $E(pUq) = \{????\}$

Function τ : $\tau(\{\text{????}\}) = \{\text{????}\}$



Fixpoint of Functions

- x is a fixpoint of τ iff $\tau(x) = x$
- $\text{lfp. } \tau$
 - a least fixpoint of τ
- $\text{gfp. } \tau$
 - a greatest fixpoint of τ

Computing lfp. τ

$$\emptyset \rightarrow \tau(\emptyset) \neq \tau(\tau(\emptyset)) \neq \tau(\tau(\tau(\emptyset))) = \tau(\tau(\tau(\tau(\emptyset))))$$

■ Compute lfp. $\tau(Y)$

- 1) $Y := \emptyset;$
- 2) **repeat**
- 3) $W := Y;$
- 4) $Y := \tau(Y)$
- 5) **until** $W = Y;$
- 6) **return** $Y;$

Computing gfp. τ

$$S \rightarrow \tau(S) \not\rightarrow \tau(\tau(S)) \not\rightarrow \tau(\tau(\tau(S))) = \tau(\tau(\tau(\tau(S))))$$

■ Compute gfp. $\tau(Y)$

- 1) $Y := S;$
- 2) **repeat**
- 3) $W := Y;$
- 4) $Y := \tau(Y)$
- 5) **until** $W = Y;$
- 6) **return** $Y;$

$$AFp = \{????\}$$

- Theorem (Clark & Emerson, 1981):
 - $AFp = \text{lfp} . \tau(Y) = \text{lfp} . [p \vee AX(Y)]$

$$\tau(Y) = p \vee AX(Y)$$

$$\emptyset \rightarrow \tau(\emptyset) \neq \tau(\tau(\emptyset)) \neq \tau(\tau(\tau(\emptyset))) = \tau(\tau(\tau(\tau(\emptyset))))$$

$$EG\ p = \{????\}$$

- Theorem (Clark & Emerson, 1981):
 - $EGp = \text{gfp} . \tau(Y) = \text{gfp} . [p \wedge EX(Y)]$

$$\tau(Y) = p \wedge EX(Y)$$

$$S \rightarrow \tau(S) \not\rightarrow \tau(\tau(S)) \not\rightarrow \tau(\tau(\tau(S))) = \tau(\tau(\tau(\tau(S))))$$

Theorem (Clark & Emerson, 1981)

- Theorem (Clark & Emerson, 1981):

- Given a finite $M=(S, R, L)$
- $AFp = \text{lfp}.\tau(Z) = \text{lfp}.[p \vee AX(Z)]$
- $EFp = \text{lfp}.\tau(Z) = \text{lfp}.[p \vee EX(Z)]$
- $AGp = \text{gfp}.\tau(Z) = \text{gfp}.[p \wedge AX(Z)]$
- $EGp = \text{gfp}.\tau(Z) = \text{gfp}.[p \wedge EX(Z)]$
- $A(pUq) = \text{lfp}.\tau(Z) = \text{lfp}.[q \vee (p \wedge AX(Z))]$
- $E(pUq) = \text{lfp}.\tau(Z) = \text{lfp}.[q \vee (p \wedge EX(Z))]$

- τ is a function: $2^S \rightarrow 2^S$
- lfp = Least Fix Point
- gfp = Greatest Fix Point

Computing lfp and gfp

- Theorem (Clark & Emerson, 1981):
- Given a finite $M=(S, R, L)$
- $AFp = \text{lfp}.\tau(Z) = \text{lfp}.[p \vee AX(Z)]$
- $EFp = \text{lfp}.\tau(Z) = \text{lfp}.[p \vee EX(Z)]$
- $AGp = \text{gfp}.\tau(Z) = \text{gfp}.[p \wedge AX(Z)]$
- $EGp = \text{gfp}.\tau(Z) = \text{gfp}.[p \wedge EX(Z)]$
- $A(pUq) = \text{lfp}.\tau(Z) = \text{lfp}.[q \vee (p \wedge AX(Z))]$
- $E(pUq) = \text{lfp}.\tau(Z) = \text{lfp}.[q \vee (p \wedge EX(Z))]$

$$\emptyset \rightarrow \tau(\emptyset) \neq \tau(\tau(\emptyset)) \neq \tau(\tau(\tau(\emptyset))) = \tau(\tau(\tau(\tau(\emptyset))))$$

$$S \rightarrow \tau(S) \neq \tau(\tau(S)) \neq \tau(\tau(\tau(S))) = \tau(\tau(\tau(\tau(S))))$$

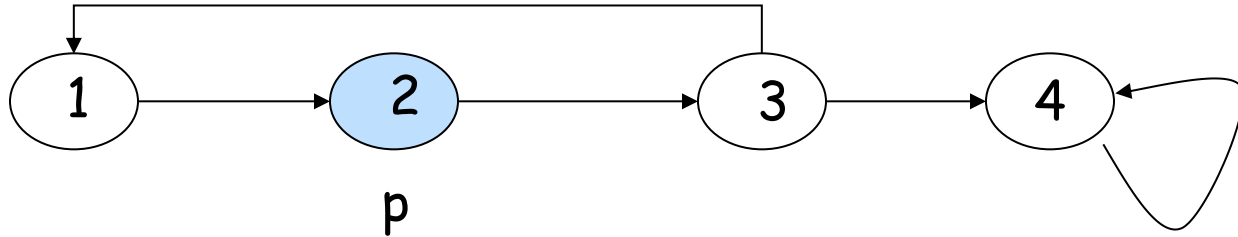
What is τ ?

- Theorem (Clark & Emerson, 1981):

- Given a finite $M=(S, R, L)$
- $AFp = \text{lfp}.\tau(Z) = \text{lfp}.[p \vee AX(Z)]$
- $EFp = \text{lfp}.\tau(Z) = \text{lfp}.[p \vee EX(Z)]$
- $AGp = \text{gfp}.\tau(Z) = \text{gfp}.[p \wedge AX(Z)]$
- $EGp = \text{gfp}.\tau(Z) = \text{gfp}.[p \wedge EX(Z)]$
- $A(p \cup q) = \text{lfp}.\tau(Z) = \text{lfp}.[q \vee (p \wedge AX(Z))]$
- $E(p \cup q) = \text{lfp}.\tau(Z) = \text{lfp}.[q \vee (p \wedge EX(Z))]$

- $AF p = p \vee AX AF p$
- $EF p = p \vee EX EF p$
- $AG p = p \wedge AX AG p$
- $EG p = p \wedge EX EG p$
- $A [p \cup q] = q \vee (p \wedge AX A [p \cup q])$
- $E [p \cup q] = q \vee (p \wedge EX E [p \cup q])$

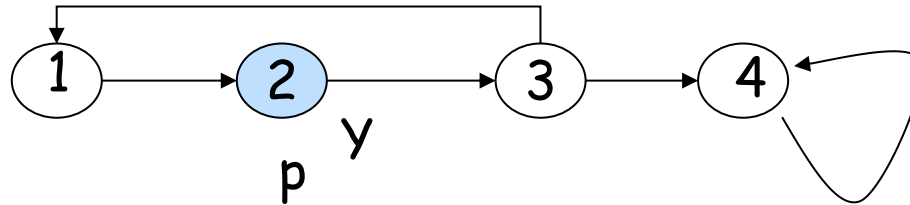
$$EFp = \text{lfp. } [p \vee EX Z] = \{ ? \}$$



$$\emptyset \rightarrow \tau(\emptyset) \not\rightarrow \tau(\tau(\emptyset)) \not\rightarrow \tau(\tau(\tau(\emptyset))) = \tau(\tau(\tau(\tau(\emptyset))))$$

- Consider EFp in the following model:
 - $|S| = 4$ and $\tau(Y) = p \vee EX(Y)$
- False does not hold in any states
 - False represents the empty set of states

$$EFp = \text{Ifp. } [p \vee EX Z]: \tau^1(\emptyset) = p$$

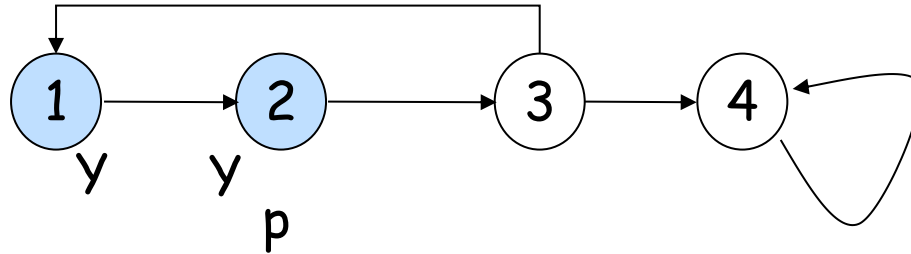


$$\tau^1(\emptyset) = p \vee EX(\emptyset) = \{2\} \cup \emptyset = \{2\} = p$$

$$\emptyset \rightarrow \tau(\emptyset) \not\rightarrow \tau(\tau(\emptyset)) \not\rightarrow \tau(\tau(\tau(\emptyset))) = \tau(\tau(\tau(\tau(\emptyset))))$$

- $EX(\emptyset) = EX(\text{False})$: set of states such that False holds in at least one of their next states
- Use γ to mark the states where the current $\tau^1(\emptyset)$ holds

$$EFp = \text{Ifp. } [p \vee EX Z]: \tau^2(\emptyset) = \{1, 2\}$$

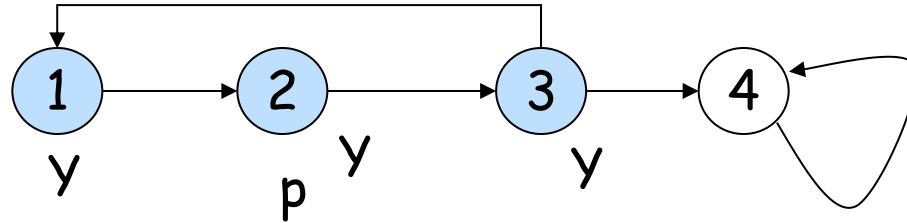


$$\tau(\emptyset) = \{2\}$$

$$\tau(\tau(\emptyset)) = \tau^2(\emptyset) = \mathbf{p} \vee \mathbf{EX}(p) = \{2\} \cup EX(\{2\}) = \{2\} \cup \{1\} = \{1, 2\}$$

$$\emptyset \rightarrow \tau(\emptyset) \not\rightarrow \tau(\tau(\emptyset)) \not\rightarrow \tau(\tau(\tau(\emptyset))) = \tau(\tau(\tau(\tau(\emptyset))))$$

Iteration 3: EFp = {1,2,3}



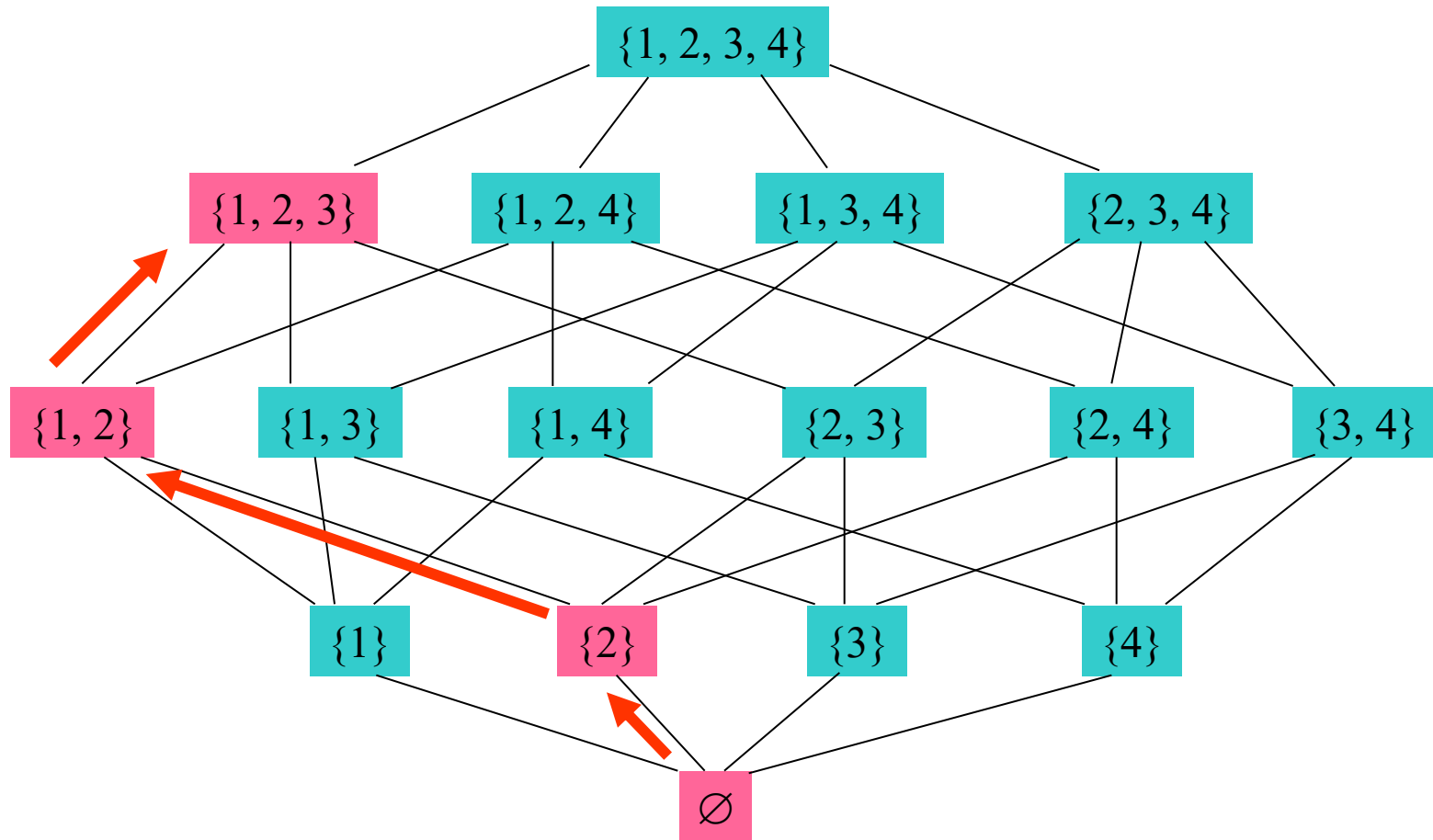
$$\begin{aligned}
 \tau(\tau(\tau(\emptyset))) &= \tau^3(\emptyset) = p \vee EX(p \vee EXp) \\
 &= \{2\} \cup EX(\{1, 2\}) \text{ /*in s1 OR s2*/} \\
 &= \{2\} \cup EX(\{1\}) \cup EX(\{2\}) \\
 &= \{2\} \cup \{3, 1\} \\
 &= \{1, 2, 3\}
 \end{aligned}$$

$$\tau^2(\emptyset) = \{1, 2\}$$

$$\emptyset \rightarrow \tau(\emptyset) \not\rightarrow \tau(\tau(\emptyset)) \not\rightarrow \tau(\tau(\tau(\emptyset))) = \tau(\tau(\tau(\tau(\emptyset))))$$

$$\text{Iteration 4: } \tau^4(\text{False}) = \tau^3(\text{False})$$

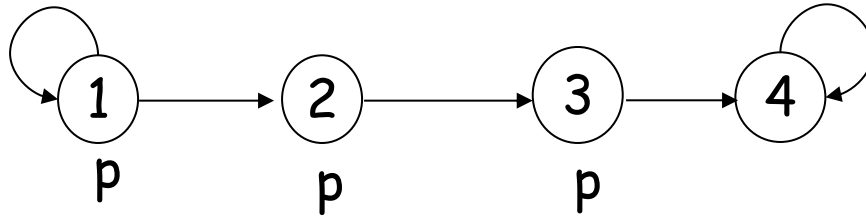
Computation Path in Lattice



$$EFp = \text{lfp. } [p \vee EX Z]$$

- Each iteration propagates the formula EFp backward in the graph by one step
- When fixpoint reached, γ labels exactly the set of states on a path to a state labeled with p
- To check if EFp holds in a certain state s , check if $s \in EFp$
- Properties characterized as least fixpoints correspond to Eventualities

$$EGp = \text{gfp. } [p \wedge EX Z] = \{ ? \}$$

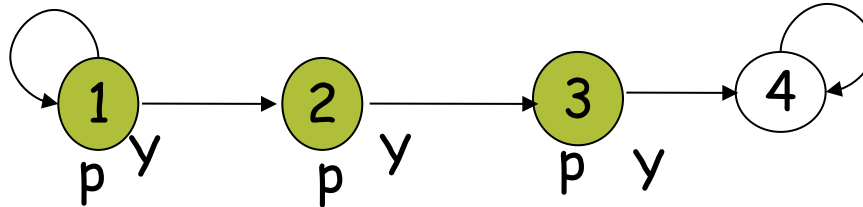


$$S = \{1, 2, 3, 4\}$$

$$S \rightarrow \tau(S) \not\rightarrow \tau(\tau(S)) \not\rightarrow \tau(\tau(\tau(S))) = \tau(\tau(\tau(\tau(S))))$$

- EGp in the following model: $|S| = 4$ and $\tau(Y) = p \wedge EX(Y)$
- True holds in all states
 - True represents the set of all states, marked by Y

$$EGp = \text{gfp. } [p \wedge EX Z]: \tau^1(S) = p$$

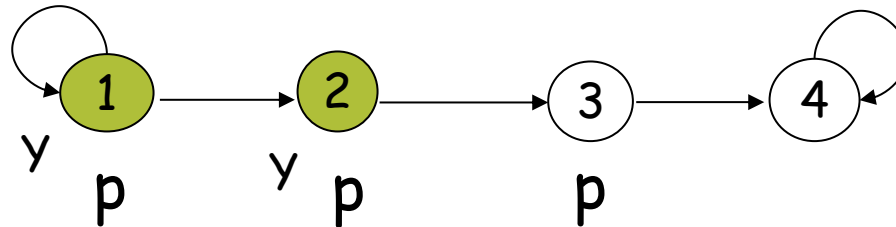


Iteration 1:

$$\tau^1(S) = p \wedge EX(S) = \{1, 2, 3\} \cap \{1, 2, 3, 4\} = \{1, 2, 3\} = p$$

$$S \rightarrow \tau(S) \not\rightarrow \tau(\tau(S)) \not\rightarrow \tau(\tau(\tau(S))) = \tau(\tau(\tau(\tau(S))))$$

$$EGp = \text{gfp. } [p \wedge EX Z]: \tau^2(S) = \{1, 2\}$$



Iteration 1:

$$\tau^1(S) = \{1, 2, 3\}$$

Iteration 2:

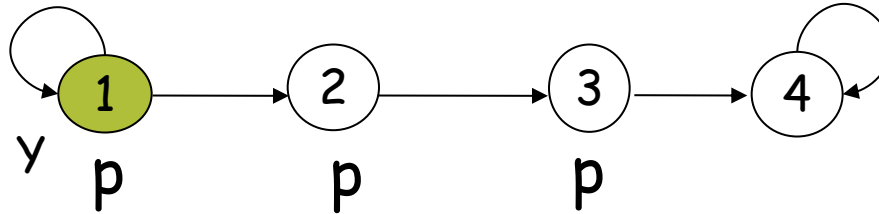
$$\tau^2(S) = p \wedge EX(p)$$

$$= \{1, 2, 3\} \cap EX\{1, 2, 3\}$$

$$= \{1, 2, 3\} \cap \{1, 2\} = \{1, 2\}$$

$$S \rightarrow \tau(S) \not\rightarrow \tau(\tau(S)) \not\rightarrow \tau(\tau(\tau(S))) = \tau(\tau(\tau(\tau(S))))$$

$$EGp = \text{gfp. } [p \wedge EX Z]: \tau^3(S) = \{1\}$$



Iteration 2:

$$\tau^2(S) = \{1, 2\}$$

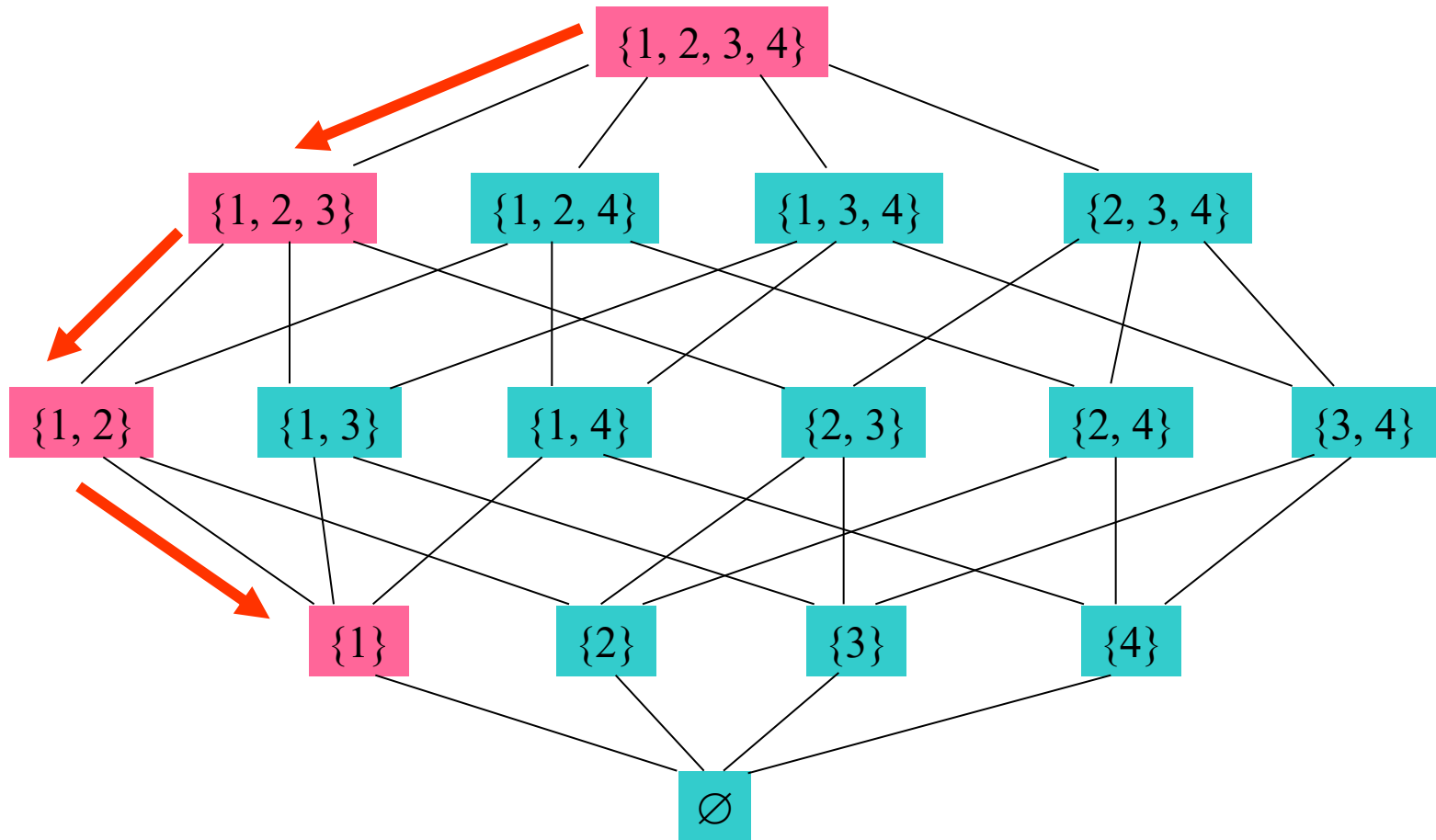
Iteration 3:

$$\begin{aligned}
 \tau^3(S) &= p \wedge EX(p \wedge EX p) \\
 &= \{1, 2, 3\} \cap EX\{1, 2\} \\
 &= \{1, 2, 3\} \cap \{1\} = \{1\}
 \end{aligned}$$

$$S \rightarrow \tau(S) \not\rightarrow \tau(\tau(S)) \not\rightarrow \tau(\tau(\tau(S))) = \tau(\tau(\tau(\tau(S))))$$

$$\text{Iteration-4: } \tau^4(\text{True}) = \tau^3(\text{True})$$

Computation Path in Lattice



Summary for EGp

- At iteration i , Y labels the set of states such that there is a path of length i where every state satisfies p
- In the fixpoint, every state in the set has a successor in the set satisfying p
- For any state in the set, there exists an infinite path where p is always true
- To verify if EGp holds in a certain state s , check if $s \in EG\ p$
- Properties characterized as greatest fixpoints correspond to Invariant properties

First Step: $EX(F)$ and $EX(T)$

- $EX(F) = F$
- $EX(\emptyset) = \emptyset$
- $AX(F) = F$
- $AX(\emptyset) = \emptyset$

- $EX(T) = T$
- $EX(S) = S$
- $AX(T) = T$
- $AX(S) = S$

- Any formula f $\Leftrightarrow$ a set S_f of states where f is true
- False $\Leftrightarrow$ the empty set $\emptyset$ ($\perp$)
- True $\Leftrightarrow$ the total set of states S (T)

CTL Model Checking

- Given a finite $M=(S, R, L)$
- $A F p = \text{lfp. } [p \vee AX(Z)]$
- $E F p = \text{lfp. } [p \vee EX(Z)]$
- $A G p = \text{gfp. } [p \wedge AX(Z)]$
- $E G p = \text{gfp. } [p \wedge EX(Z)]$
- $A (p U q) = \text{lfp. } [q \vee (p \wedge AX(Z))]$
- $E (p U q) = \text{lfp. } [q \vee (p \wedge EX(Z))]$

$H(a) = \{s \mid s \text{ is labeled with } a\}$ for atomic formula a

$H(\neg f) = S - H(f)$

$H(f \wedge g) = H(f) \cap H(g)$

$H(AXf) = \{s \mid \forall t. (s, t) \in R \Rightarrow t \in H(f)\}$

$H(EXf) = H(\neg(AX(\neg f)))$

$H(AGf) = \text{gfp. } [f \wedge AXZ] = \nu Z. (H(f) \cap \{s \mid \forall t. (s, t) \in R \Rightarrow t \in Z\})$

$H(EGf) = H(\neg(AG\neg f))$

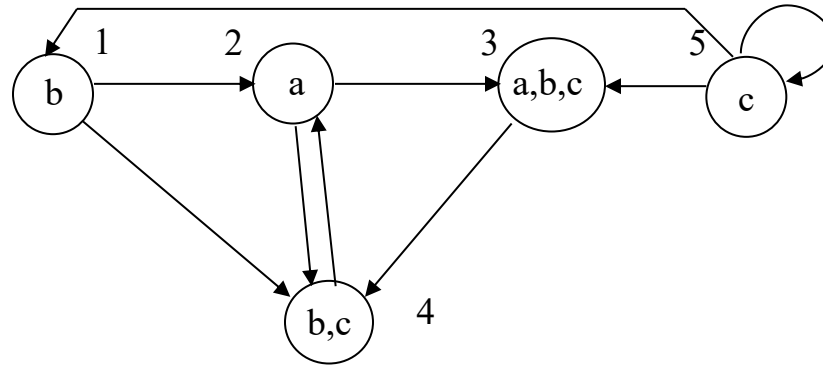
$H(AFf) = \text{lfp. } [f \vee AXZ] = \mu Z. (H(f) \cup \{s \mid \forall t. (s, t) \in R \Rightarrow t \in Z\})$

$H(EFf) = H(\neg(AG\neg f))$

$H(A(fUg)) = \text{lfp. } [g \vee (f \wedge AXZ)] = \mu Z. (H(g) \cup (H(f) \cap \{s \mid \forall t. (s, t) \in R \Rightarrow t \in Z\}))$

$H(E(fUg)) = H(\neg(A(\neg g U (\neg f \wedge \neg g)) \vee AG(\neg g)))$

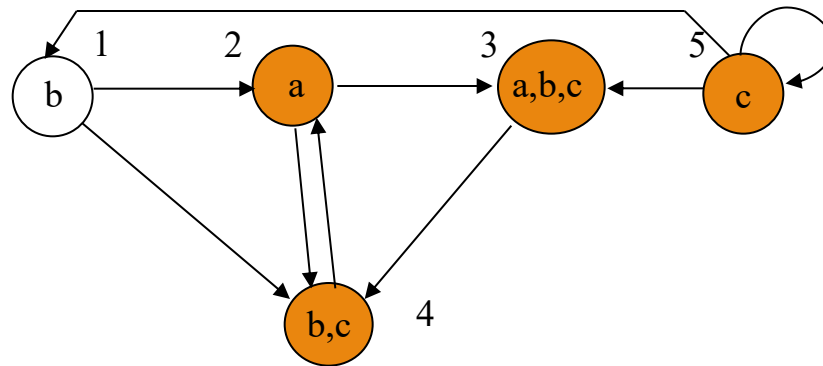
Example



Structure M:

- $S = \{1, 2, 3, 4, 5\}$, $AP = \{a, b, c\}$,
- $R = \{(1, 2), (2, 3), (5, 3), (5, 5), (5, 1), (2, 4), (4, 2), (1, 4), (3, 4)\}$
- $L(1) = \{b\}$, $L(2) = \{a\}$, $L(3) = \{a, b, c\}$, $L(4) = \{b, c\}$, $L(5) = \{c\}$

Property: $AG(a \vee c)$



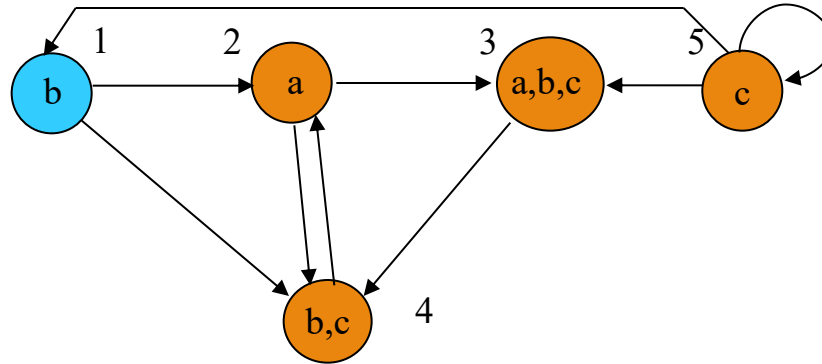
$$\begin{aligned} H(a \vee c) &= H(a) \cup H(c) \\ &= \{2, 3\} \cup \{3, 4, 5\} \\ &= \{2, 3, 4, 5\} \end{aligned}$$

$$\begin{aligned} \text{Let } p &= a \vee c \\ H(p) &= \{2, 3, 4, 5\} \\ AG(a \vee c) &= AGp \end{aligned}$$

$$AGp = \text{gfp. } [p \wedge AX Z]$$

- Let $p = a \vee c$
- $AGp = \text{gfp. } [p \wedge AX Z]$
- $H(p) = \{2, 3, 4, 5\}$
- $AX Z = \{s \mid \forall t. (s, t) \in R \Rightarrow t \in Z\}$
- $H(AG(p)) = \text{gfp. } \{2, 3, 4, 5\} \cap \{s \mid \forall t. (s, t) \in R \Rightarrow t \in Z\}$

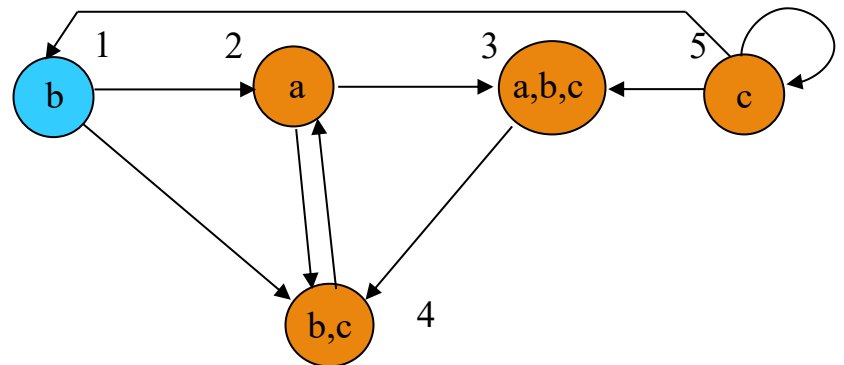
$$AGp = \text{gfp. } [p \wedge AX Z]$$



- $H(AG(p)) = \text{gfp. } \{2, 3, 4, 5\} \cap \{s \mid \forall t. (s, t) \in R \Rightarrow t \in Z\}$
- $Z_0 = S = \{1, 2, 3, 4, 5\} = \text{True}$
- $Z_1 = \{2, 3, 4, 5\} \cap \{s \mid \forall t. (s, t) \in R \Rightarrow t \in Z_0\}$
 $= \{2, 3, 4, 5\} \cap \{1, 2, 3, 4, 5\}$
 $= \{2, 3, 4, 5\}$

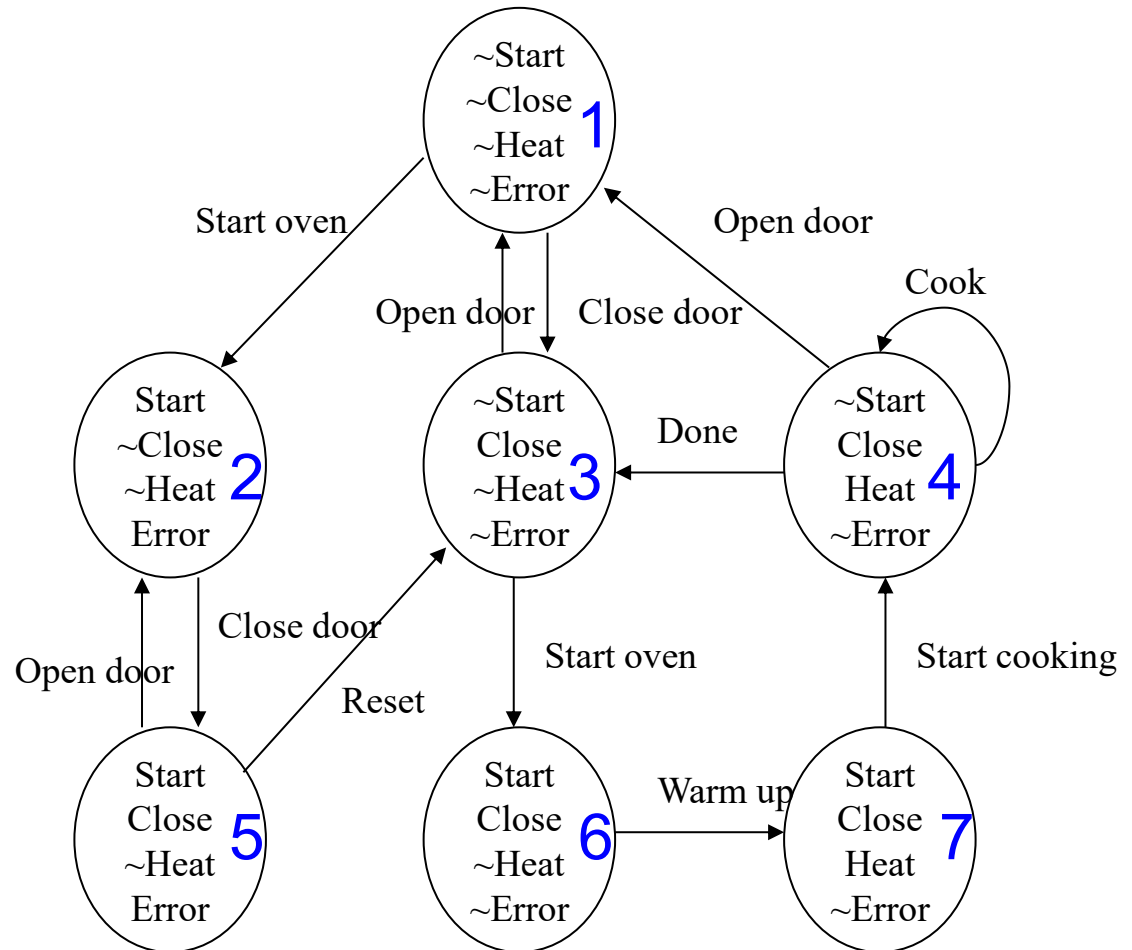
$$AGp = \text{gfp. } [p \wedge AX Z]$$

- $Z_1 = \{2, 3, 4, 5\}$
- $Z_2 = \{2, 3, 4, 5\} \cap \{s \mid \forall t. (s, t) \in R \Rightarrow t \in Z_1\}$
 $= \{2, 3, 4, 5\} \cap \{1, 2, 3, 4\}$
 $= \{2, 3, 4\}$
- $Z_3 = \{2, 3, 4, 5\} \cap \{s \mid \forall t. (s, t) \in R \Rightarrow t \in Z_2\}$
 $= \{2, 3, 4, 5\} \cap \{1, 2, 3, 4\}$
 $= \{2, 3, 4\}$
- $Z_3 = Z_2$

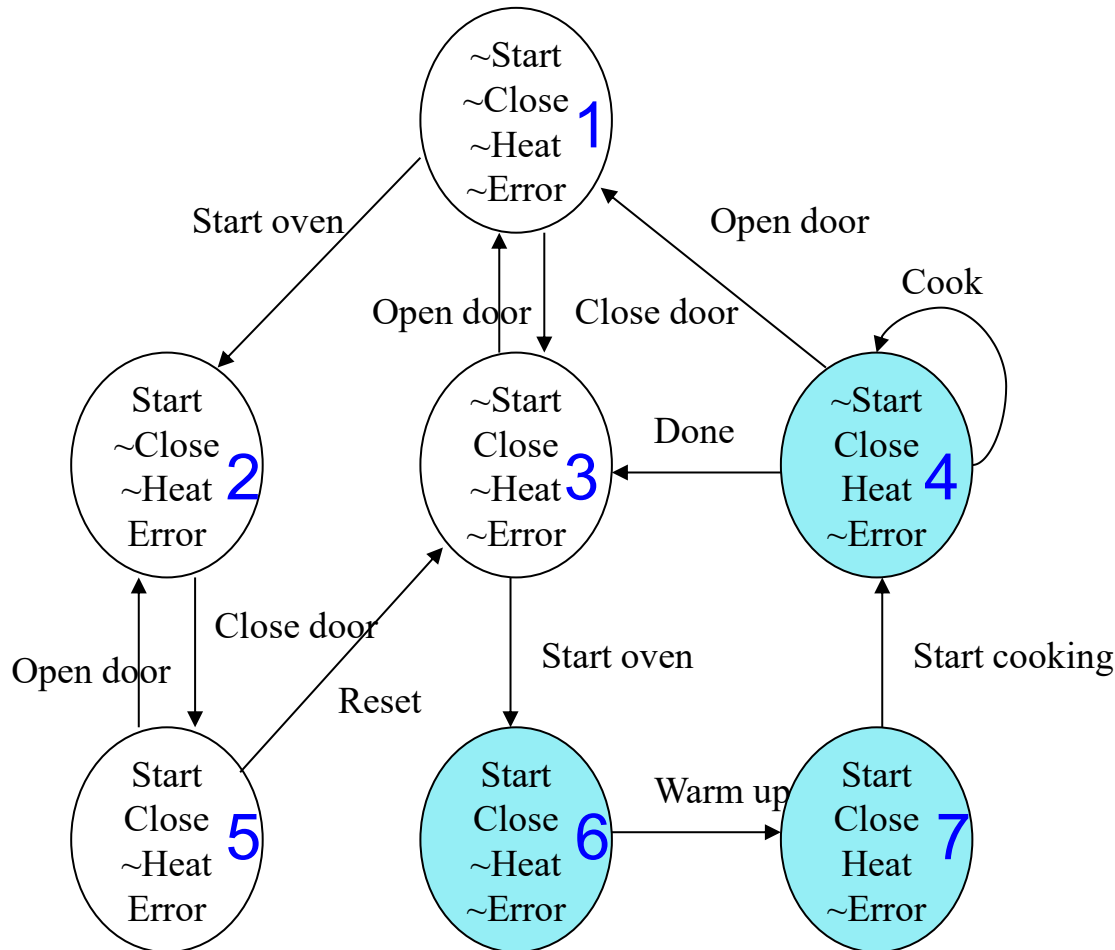


AG (Start $\rightarrow$ AF Heat)

- Can you try the fixpoint method?

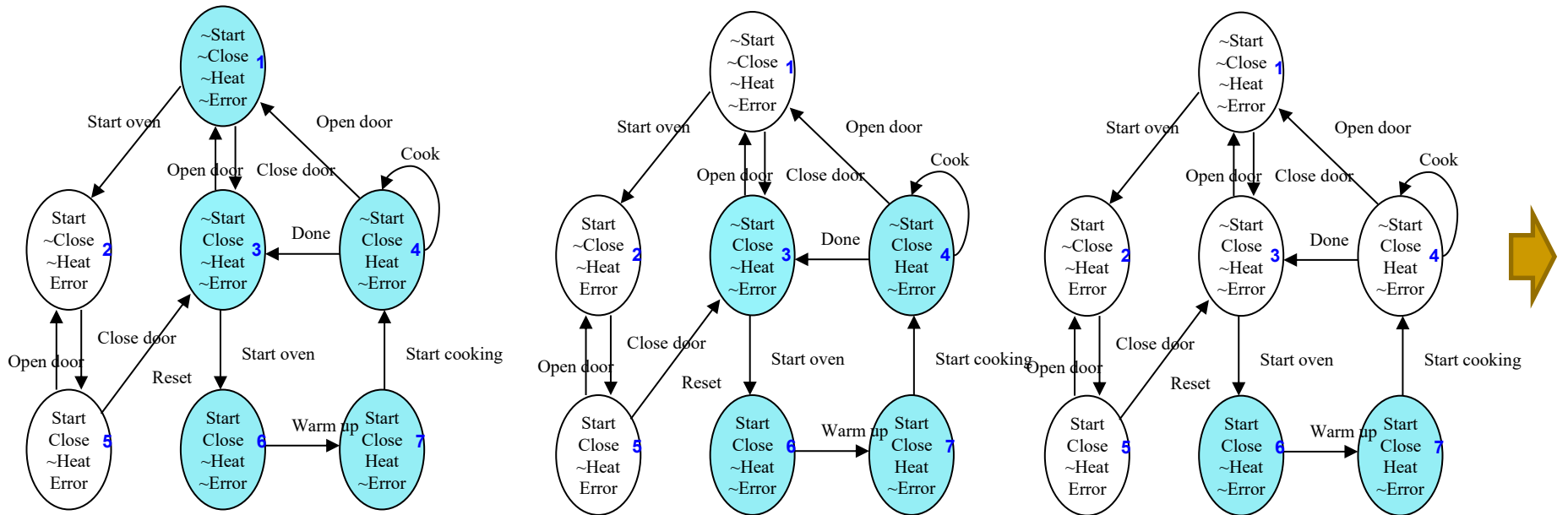


$$AG(\text{Start} \rightarrow AF\text{Heat}) = \emptyset$$



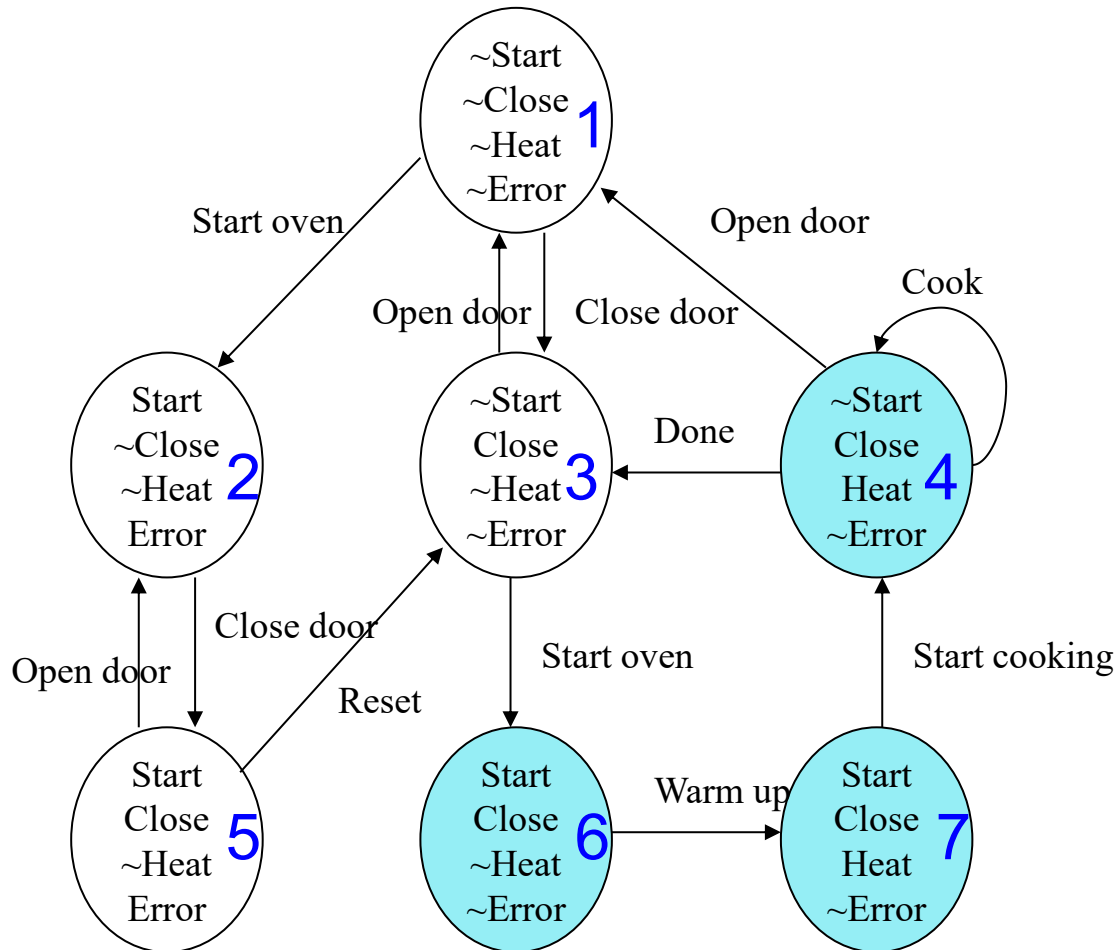
- $AG(\text{Start} \rightarrow AF\text{Heat})$
- $= AG(\sim\text{Start} + AF\text{Heat})$
- $AF\text{Heat} = \text{lfp}[\text{Heat} \vee AX(Z)]$
- $\tau(Z) = \text{Heat} \vee AX(Z)$
- $\tau(F) = \{4, 7\},$
- $\tau^2(F) = \{4, 6, 7\},$
- $\tau^3(F) = \{4, 6, 7\}, \dots$

$$AG(\text{Start} \rightarrow AF\text{Heat}) = \emptyset$$



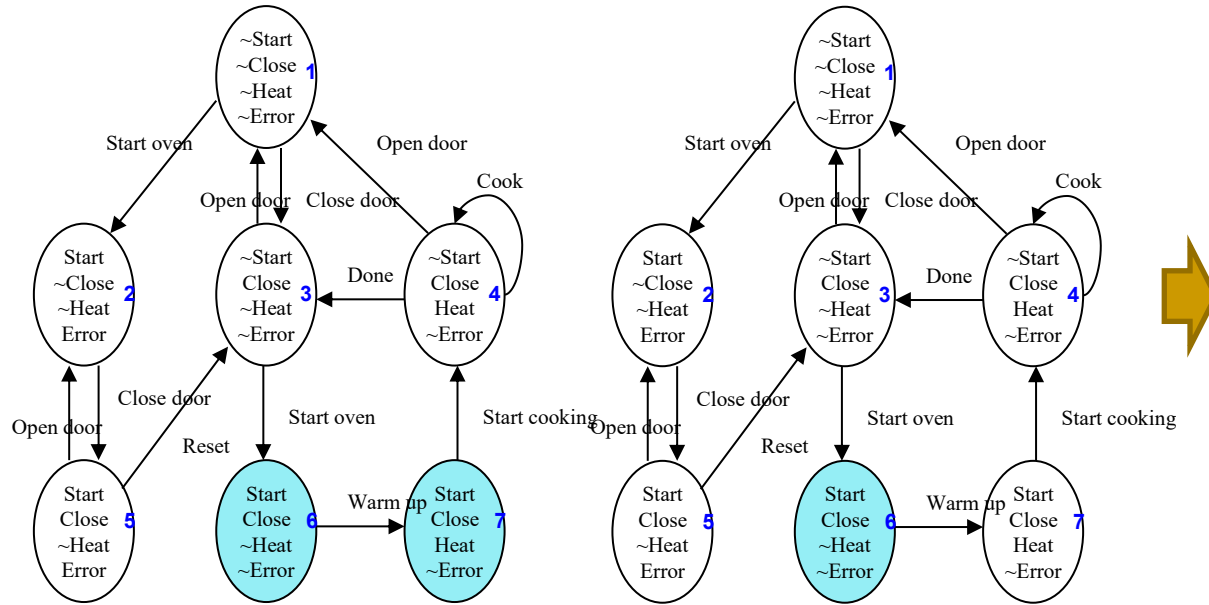
- $p = (\sim\text{Start} + AF\text{Heat}) = \{1, 3, 4\} \cup \{4, 6, 7\} = \{1, 3, 4, 6, 7\}$
- $AG(\sim\text{Start} + AF\text{Heat}) = AG(p) = \text{gfp}. [p \wedge AX(Z)]$
- $\tau(Z) = p \wedge AX(Z)$
- $\tau(T) = p, \tau^2(T) = \{3, 4, 6, 7\}, \tau^3(T) = \{6, 7\}, \tau^4(T) = \{6\}, \tau^5(T) = \{ \}, \dots$

$AG(\text{Start} \wedge AF\text{Heat}) = ?$



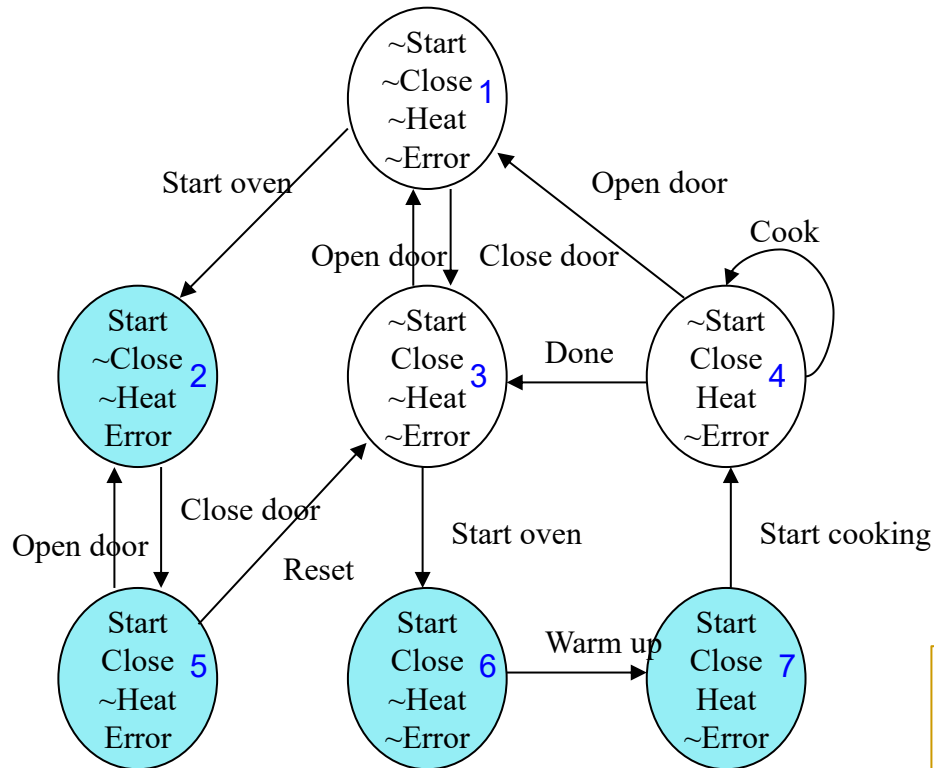
- $AG(\text{Start} \wedge AF\text{Heat})$
- $AF\text{Heat} = \text{lfp}[\text{Heat} \vee AX(Z)]$
- $\tau(Z) = \text{Heat} \vee AX(Z)$
- $\tau(F) = \{4, 7\},$
- $\tau^2(F) = \{4, 6, 7\},$
- $\tau^3(F) = \{4, 6, 7\}, \dots$

$$AG(\text{Start} \wedge AF\text{Heat}) = \emptyset$$



- $p = (\text{Start} \wedge AF\text{Heat}) = \{2, 5, 6, 7\} \cap \{4, 6, 7\} = \{6, 7\}$
- $AG(\text{Start} \wedge AF\text{Heat}) = AG(p) = \text{gfp}. [p \wedge AX(Z)]$
- $\tau(Z) = p \wedge AX(Z)$
- $\tau(T) = p, \tau^2(T) = \{6\}, \tau^3(T) = \{ \}, \dots$

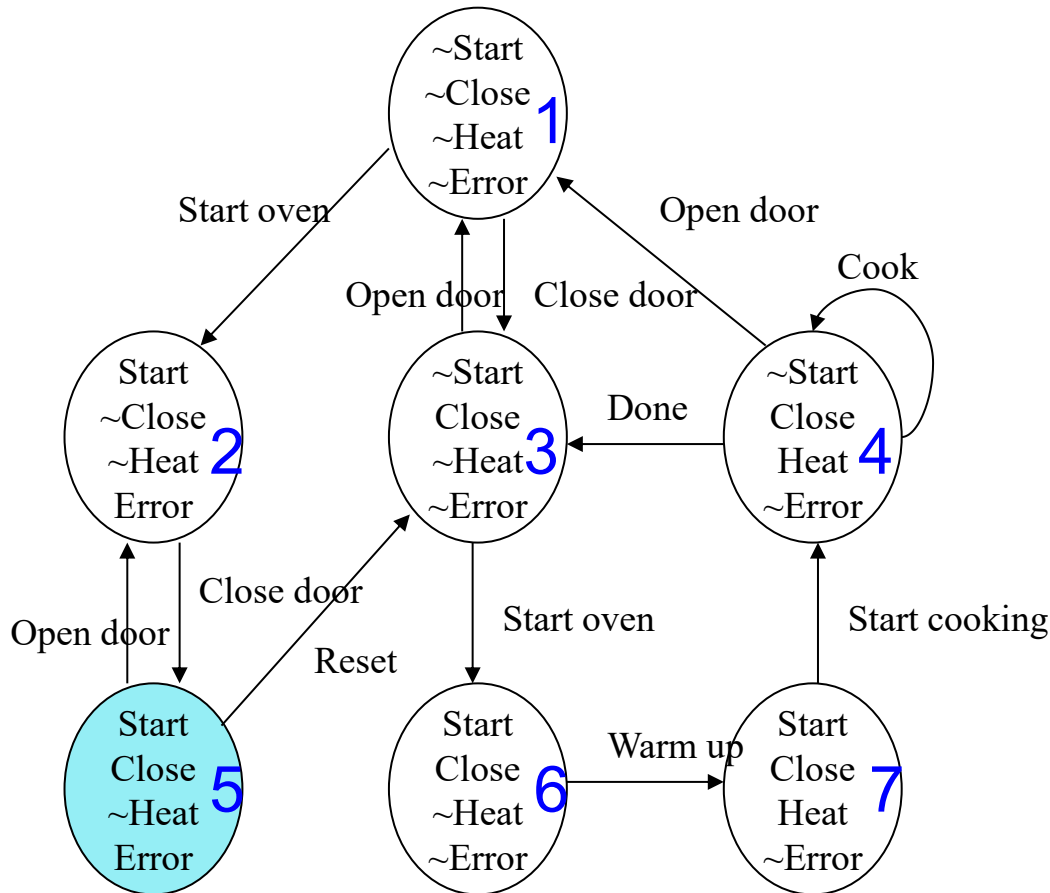
Comparison: $\rightarrow$ and $\wedge$



- $AG(Start \rightarrow AFHeat) = \emptyset$
- If $Start=1$, then $AFHeat=1$
- If $Start=0$,
 - then $(Start \rightarrow AFHeat)=1$

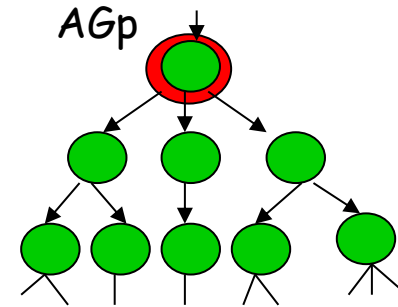
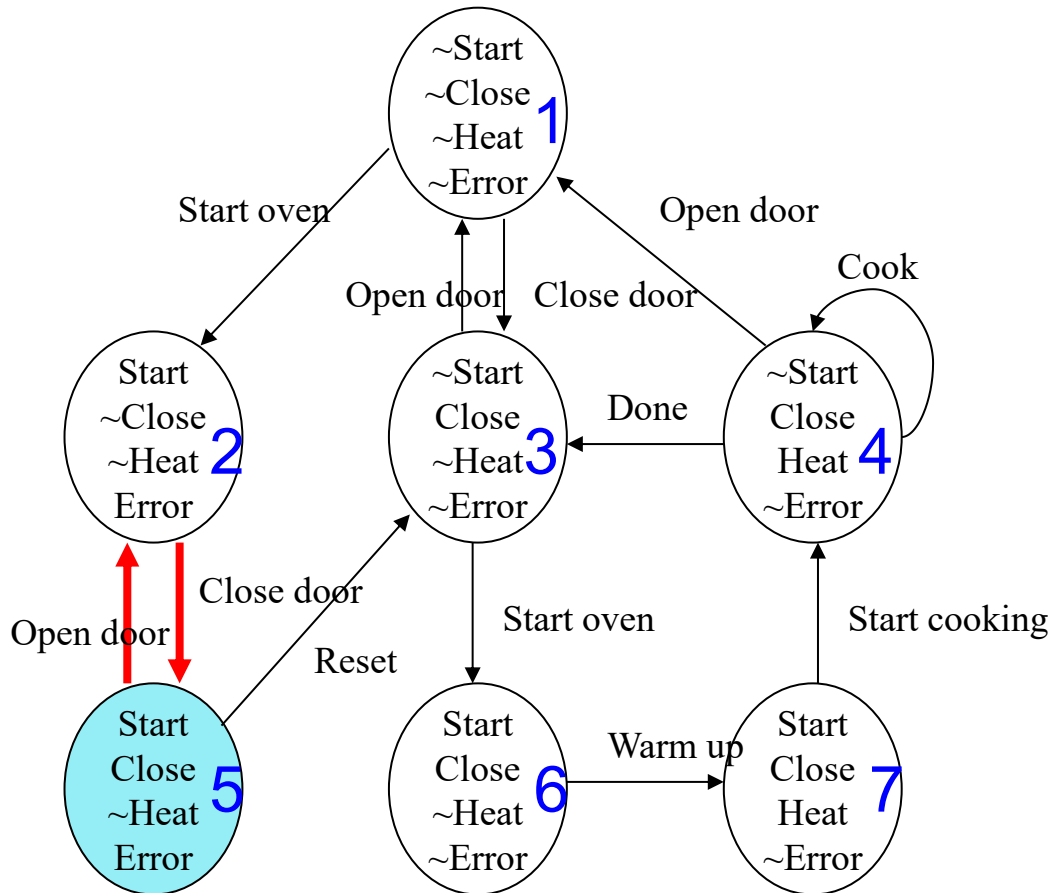
- $AG(Start \wedge AFHeat) = \emptyset$
- We require that $Start=1$ be everywhere !

$AG((Start \wedge Closed) \rightarrow AFHeat) = ?$



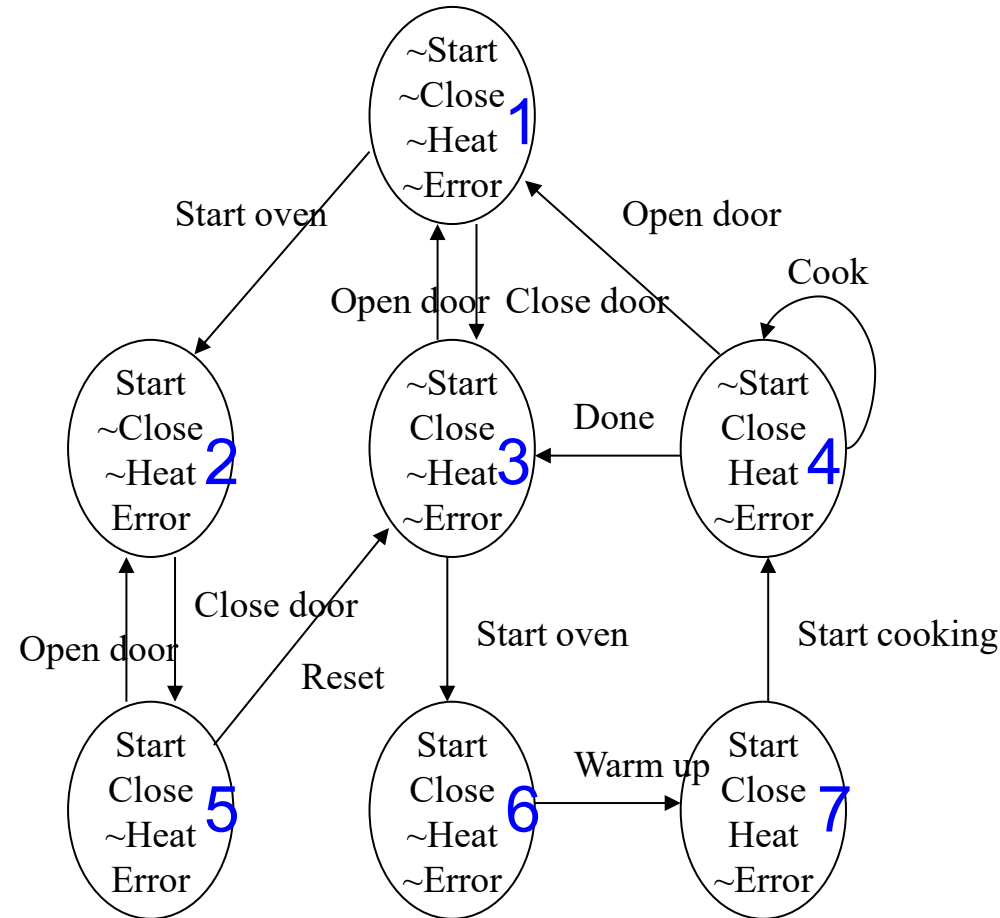
- $AFH = \{4, 6, 7\},$
- $AG(S \wedge Close \rightarrow AFH)$
- $= AG((S \wedge Close) \rightarrow AFH)$
- $= AG(\neg(S \wedge Close) + AFH)$
- $= AG(\neg S + \neg Close + AFH)$
- $= AG(\{1, 3, 4\} + \{1, 2\} + \{4, 6, 7\})$
- $= AG(\{1, 2, 3, 4, 6, 7\}) = AGp$
- $\tau(Z) = p \wedge AX(Z)$
- $\tau(T) = p, \tau^2(T) = \{1, 3, 4, 6, 7\},$
- $\tau^3(T) = \{3, 4, 6, 7\},$
- $\tau^4(T) = \{6, 7\},$
- $\tau^5(T) = \{6\}, \tau^6(T) = \{ \}.$

$AG((Start \wedge Close) \rightarrow AFHeat) = \emptyset$, Why?



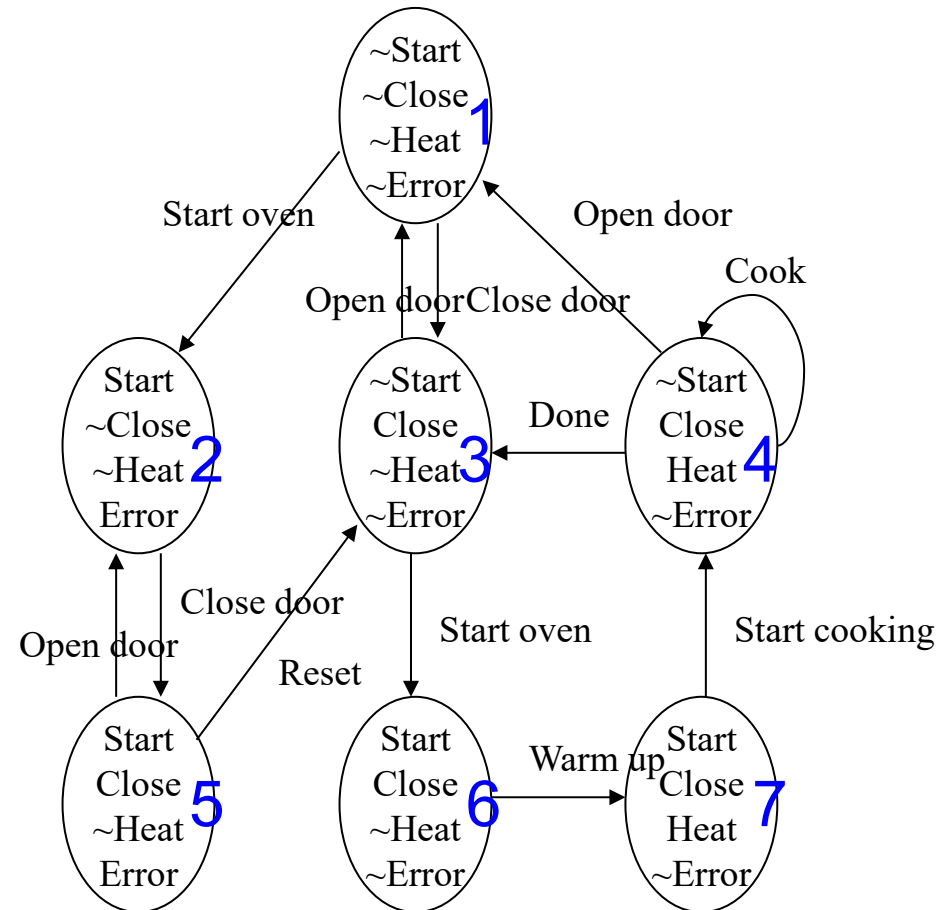
- Every state can reach all the other states.
- If some node satisfies AGp , p is true in all the other nodes.
- At state 5, $Start \wedge Close = true$, however, $AFHeat$ is not true due to the infinite path: $5 \rightarrow 2 \rightarrow 5 \rightarrow 2 \rightarrow \dots$

$AG((Start \wedge Closed) \rightarrow EFHeat) = ?$



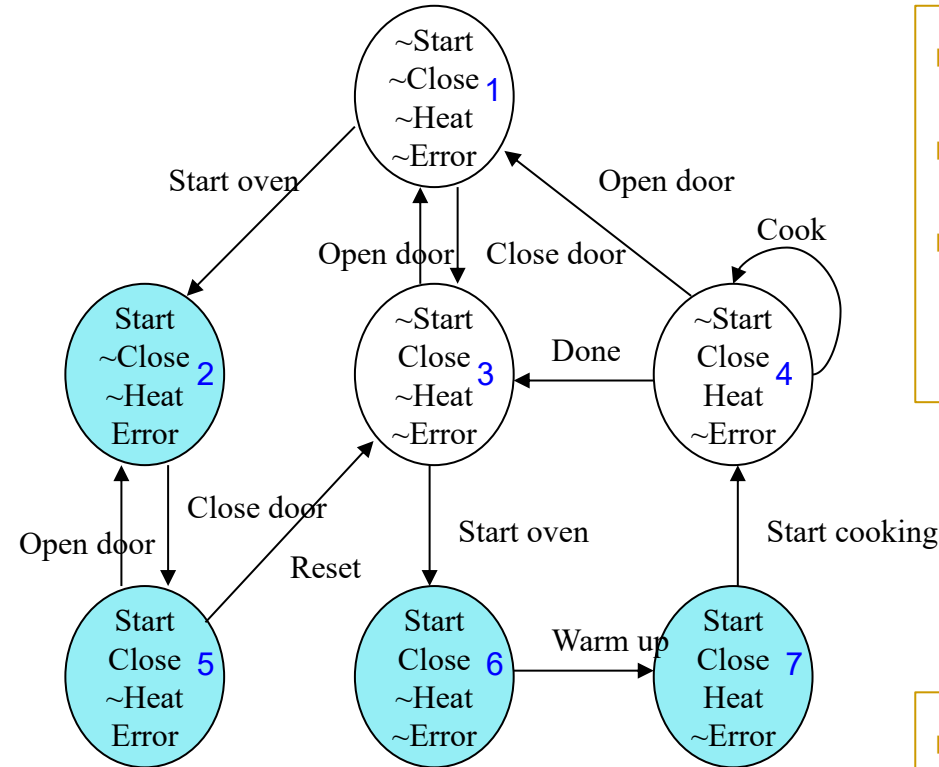
- $EFH = T$
- $AG(S \wedge Close \rightarrow EFH)$
- $= AG((S \wedge Close) \rightarrow EFH)$
- $= AG(\neg(S \wedge Close) + EFH)$
- $= AG(\neg S + \neg Close + EFH)$
- $= AG(\{1,3,4\} + \{1,2\} + T)$
- $= AG(T) = T$

$$AG((Start \wedge Closed) \wedge EFHeat) = ?$$



- $EFH = T$
- $AG(S \wedge Close \wedge EFH)$
- $= AG(\{2,5,6,7\} \cap \{3,4,5,6,7\} \cap T)$
- $= AG(\{5,6,7\})$
- $= \tau(Z) = p \wedge AX(Z)$
- $\tau(T) = p, \tau^2(T) = \{2,6\}, \tau^3(T) = \{ \}, \dots$

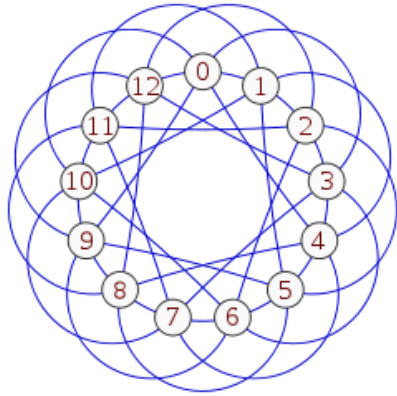
Comparison: $\rightarrow$ and $\wedge$



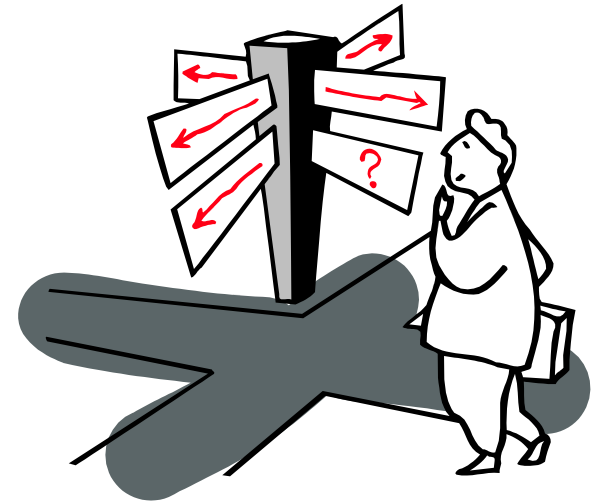
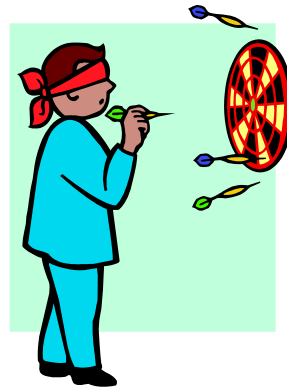
- $AG((Start \wedge Closed) \rightarrow EFHeat) = T$
- If $Start \wedge Closed = 1$, then $EFHeat = 1$
- If $Start \wedge Closed = 0$,
 - then $(Start \wedge Closed \rightarrow AFHeat) = 1$

- $AG((Start \wedge Closed) \wedge EFHeat) = \emptyset$
- We require that $Start \wedge Closed = 1$ be everywhere !

Summary: Model Checking



, {??????} |= Property



References

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- E. M. Clark, O. Grumberg and D. A. Peled, *"Model Checking"*, MIT Press, 1999.
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